

Axel Lojas

ELECTRICAL ENGINEERING STUDENT · ELECTRICAL DESIGN · PROJECT ENGINEERING

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Summary

Electrical Engineering student with interests in power systems, control systems, and electrical design. Experienced in electrical design, controls, testing, and technical analysis, with proficiency in MATLAB, Python, computer vision, and engineering simulation tools. Seeking opportunities to apply technical and problem-solving skills to electrical construction, power distribution, and project engineering in complex urban environments.

Work Experience

Brookhaven National Laboratory · Department of Energy · Research Foundation CUNY

Upton, New York

ACCELERATOR TESTING FACILITY INTERN

Jun. 2026 - Aug. 2026

- Characterized an IQ modulator through automated voltage sweeps and RF measurements to support replacement of an aging accelerator control system.
- Developed Python-based graphical interfaces for real-time equipment control, monitoring, and data acquisition.
- Designed and integrated an automated laser beam alignment system using computer vision and motorized mirror mounts, improving repeatability and reducing manual alignment.
- Developed and validated PI control algorithms for automated positioning, improving system stability and alignment accuracy.
- Tested, troubleshooted, and integrated laboratory hardware including power supplies, instrumentation, optical components, and control systems.

CUNY - The City College of New York

New York, New York

TEACHING ASSISTANT · MATH TUTOR

Mar. 2025 - Present

- Tutor undergraduate students in Algebra through Differential Equations during one-on-one sessions and recitations.
- Assist with grading, classroom instruction, and development of instructional materials.
- Guide students through analytical problem-solving using collaborative and step-by-step approaches.

Education

CUNY - The City College of New York

New York

B.S. IN ELECTRICAL ENGINEERING

January 2024 - Present

- Expected B.S. — Spring 2028
- Expected M.S. — Spring 2029
- 3.5 GPA

Skills

Engineering Software	AutoCAD, LTspice, MultiSIM, MATLAB, LabVIEW, 3DOptix, OPERA SIMULIA, Trame, Rat-GUI
Programming	Python, MATLAB, C/C++
Hands-On Skills	Electrical Troubleshooting, Construction, Welding, Autobody Repair, Vehicle Maintenance, Woodworking
Control Systems	PID Control, Computer Vision, Instrument automation
Test Equipment	Oscilloscope, Power Supplies, Laser Alignment, Optical Instrumentation
Operating Systems	Linux, Windows
Languages	Spanish, English

Extracurricular Activity

LAESA-SHPE (Latin American Engineering Student Association / Society of Hispanic Professional Engineers)

New York, New York

CO-DIRECTOR OF PRE-COLLEGE ENGINEERING DAY

Jun. 2026 - Present

- Collaborated with industry sponsors to secure funding and resources for Pre-College Engineering Day.
 - Coordinated with university departments and local schools to organize engineering outreach events.
 - Led hands-on engineering activities introducing Highschool students to fundamental engineering concepts.
- Co-Director of Pre-College Engineering Day LAESA-SHPE (Latin American Engineering Student Association / Society of Hispanic Professional Engineers) New York, New York Jun. 2026 - Present
- Collaborated with industry sponsors to secure funding and resources for Pre-College Engineering Day.
 - Coordinated with university departments and local schools to organize engineering outreach events.
 - Led hands-on engineering activities introducing Highschool students to fundamental engineering concepts.